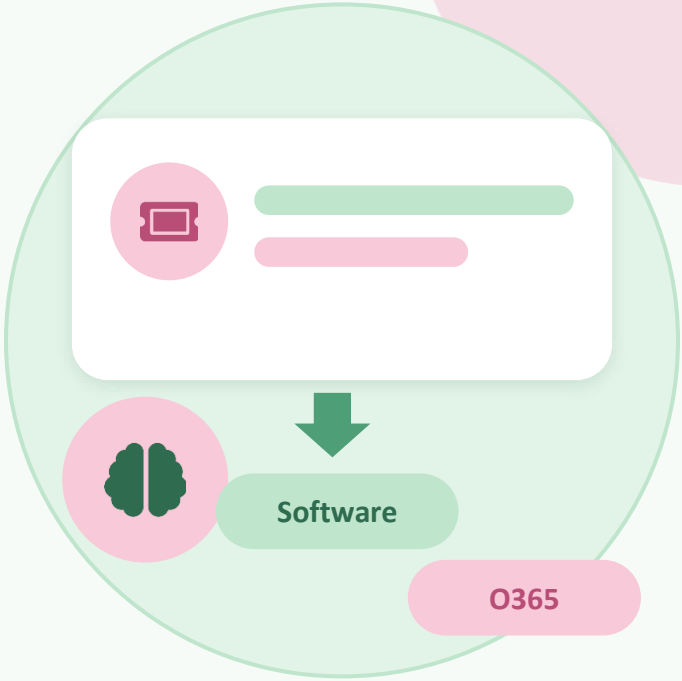
IT Helpdesk Support Ticket Classifier

Automating IT Support with Machine Learning



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Problem Statement



The Challenge

IT helpdesk teams receive high volumes of support requests daily.



The Bottleneck

Manually reading and categorizing tickets takes significant time and delays response times.



The Solution

Use Machine Learning to classify tickets from their text, enabling faster routing and improved efficiency.

Objectives

- Automatically classify incoming IT support tickets into predefined categories
- Preprocess and clean ticket text data (removing PII and noise)
- Convert text descriptions into numerical features using TF-IDF
- Train a robust classifier using Logistic Regression
- Provide a user-friendly interface to predict new tickets in real time

Methodology

01

Dataset Collection

2,229 manually classified tickets

02

Text Preprocessing

Lowercasing, punctuation removal

03

Feature Extraction

TF-IDF Vectorization

04

Model Training

Logistic Regression

05

Prediction

Classifying new text

06

Evaluation

Accuracy, Precision, Recall, F1-score

Technologies Used

Python

Core programming language

Pandas & NumPy

Data manipulation and numerical operations

Scikit-learn

TF-IDF and Logistic Regression

Matplotlib & Seaborn

Data visualization and confusion matrix

Streamlit

Interactive web application

Joblib

Saving and loading trained models

Dataset Information

Source

“Classification of IT Support Tickets” (Zenodo, public dataset)

2,229

Tickets

7

Categories

Ticket Categories

- Support general
- Fileservice
- Software
- O365
- Active Directory
- Computer-Services
- EOL

Note: text was pre-anonymized to remove Personal Identifiable Information (PII) before model processing.

Model Evaluation Metrics

Logistic Regression performance on the test set

84.30%
Accuracy

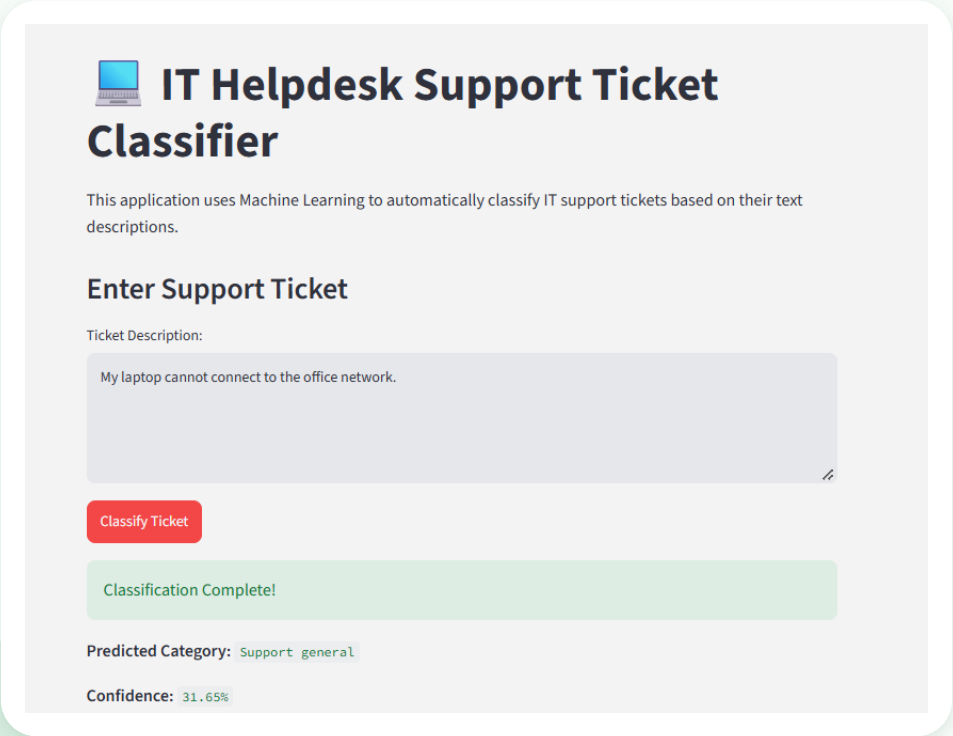
86.54%
Precision

84.30%
Recall

83.16%
F1-score

The model shows strong performance, particularly in distinguishing high-volume categories like ‘Fileservice’ and ‘Support general’.

Application Demonstration



- Interactive **Streamlit** application
- Users enter ticket descriptions directly
- Returns the **Predicted Category**
- Displays the model's **Confidence Score**

Example: "My laptop cannot connect to the office network." → Support general (31.65%)

Conclusion & Future Work

Conclusion

Machine Learning can successfully automate IT support ticket classification with high accuracy, reducing manual workload and improving response times.

Future Work

- Test more complex models such as Random Forests or Neural Networks/BERT
- Improve accuracy on complex text
- Integrate the model directly into IT Service Management tools such as Jira or ServiceNow via API